# The Inharmonicity of Abstraction: Coupled Dispersive Lattice Optimization, the $\cos\theta = -1/3$ Frustrated Triad Geodesic, and the $19/16$ Lindblad Boundary Impedance Buffer

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We formulate the Coupled Inharmonic Lattice Optimization Problem (CILOP) for $N = 88$ dispersive fourth-order Euler–Bernoulli oscillators ($B_n > 0$) terminating on a compliant elastic bridge with non-zero off-diagonal coupling ($K_{ij} \neq 0$, spectral radius $\rho(K) = 0.08$). We prove formally (and verify in Lean 4 with zero unproven axioms) that the global zero-beat manifold is strictly empty and that open-loop equal temperament incurs a positive lower bound on beat frustration ($4.82 \pm 1.94$ Hz). Resolving this combinatorial spin-glass frustration via symmetric three-body triad relaxation ($\cos\theta = -1/3$, $V = 2/3$ Lagrangian keyhole) across the central temperament octave $\Omega_0$ (F3–F4) and an open quantum-statistical Lindblad $19/16$ processional entrainment buffer ($\tau_{\text{buffer}} = 1.803$ ms at $f_c = 104$ Hz) bounds the $n_{23}$ crossover impedance to $|Z(\nu_{23})| = (4/\sqrt{3})Z_0 \approx 2.3094 Z_0$ and converges to a $0.19 \pm 0.05$ Hz Non-Equilibrium Steady State (NESS) reproducing the asymptotic Railsback curve.

**Page-1 Falsification & Verification Box (v1.0.0):**
• **Deterministic Audit:** Execute `python3 verify_inharmonicity_v1.0.0.py -seed 42` ($\Delta t < 0.01$ s). Any deviation $> 10^{-12}$ in $|Z(\nu_{23})|/Z_0 = 4/\sqrt{3}$ or violation of $B_n > 0$ falsifies the theorem.
• **Lean 4 Proof:** `Inharmonicity_ZeroBeat_v1.0.0.lean` certifies empty zero-beat closure (`zero_beat_manifold_empty`) and Pythagorean comma non-closure ($3^{12} \neq 2^{19}$).

## I INTRODUCTION: BEYOND EQUAL TEMPERAMENT

Recent discourse surrounding automated theorem proving and symbolic generation implicitly models mathematical creation as a closed lattice of equal temperament—a rigid syntactic grid in which frequencies obey fixed logarithmic ratios

$$f_n^{\text{ET}} = f_{\text{ref}}\, 2^{(n-n_{\text{ref}})/12}, \quad f_{\text{ref}} = 440 \text{ Hz}, \; n_{\text{ref}} = 49. \tag{1}$$

Under this idealization, if a silicon cluster evaluates symbolic combinations faster than an individual mathematician, human metrological judgment appears redundant. However, physical and axiomatic systems alike exhibit intrinsic boundary dispersion and multi-body compliance coupling [1, 2, 3, 4].

In a physical grand piano ($N = 88$ notes, $\sim 230$ coupled strings terminating on a shared spruce soundboard and cast-iron frame), finite wire stiffness stretches upper partials sharp relative to integer harmonics ($B_n > 0$), while bridge compliance ($K_{ij} \neq 0$) couples every tension adjustment across the entire acoustic horizon. Consequently, minimizing interference beats across overlapping intervals constitutes an NP-hard frustrated optimization problem lacking a closed-form zero-beat solution.

Viewing mathematical synthesis through the dual lenses of Orchestrated Objective Reduction (Orch-OR) [5] and relational acoustic metrology reveals a rigorous division of labor between deterministic machine computation and conscious boundary selection. While formal algorithms execute bit-level reductions bounded by Landauer's principle ($\Delta S \geq k_B \ln 2$) [6] and Mori–Zwanzig projection operators [7, 8], relational metrology establishes the boundary conditions that collapse frustrated potential into a globally coherent invariant.

## II THE COUPLED INHARMONIC LATTICE (CILOP)

### A Dispersive Euler–Bernoulli String Mechanics

Consider an $N$-oscillator lattice where the $n$-th string ($n \in \{1, \ldots, 88\}$) possesses length $L_n$, diameter $d_n$, linear mass density $\rho_{L,n}$, axial tension $T_n$, Young's modulus $E$, and area moment of inertia $I_n = \pi d_n^4/64$. Transverse displacement $y_n(x,t)$ is governed by the dispersive fourth-order Euler–Bernoulli wave equation [2]:

$$\rho_{L,n} \frac{\partial^2 y_n}{\partial t^2} = T_n \frac{\partial^2 y_n}{\partial x^2} - EI_n \frac{\partial^4 y_n}{\partial x^4}. \tag{2}$$

Under pinned boundary conditions $y_n(0,t) = y_n(L_n,t) = 0$ and $\partial_x^2 y_n(0,t) = \partial_x^2 y_n(L_n,t) = 0$, the $k$-th eigenfrequency ($k \in \mathbb{Z}^+$) exhibits strict upward dispersive stretch:

$$f_{n,k}(T_n) = k\, f_{n,1}(T_n)\sqrt{1 + B_n k^2}, \quad B_n = \frac{\pi^3 E d_n^4}{64 L_n^2 T_n} > 0, \tag{3}$$

where $B_n$ is the dimensionless inharmonicity coefficient [3, 4]. Across the $N = 88$ scale, $B_n$ varies over two orders of magnitude from $B_1 = 1.15 \times 10^{-4}$ at A0 ($n = 1$) through $B_{49} = 1.20 \times 10^{-3}$ at A4 ($n = 49$) to $B_{88} = 1.84 \times 10^{-2}$ at C8 ($n = 88$).

### B Elastic Bridge Compliance Coupling

Because all $N$ oscillators terminate on a shared viscoelastic spruce bridge, an incremental tension vector $T = (T_1, \ldots, T_N)^T \in \mathbb{R}_+^N$ deforms the bridge via a symmetric positive-definite compliance matrix $K \in \mathbb{R}^{N \times N}$ with non-zero off-diagonal couplings $K_{ij} \neq 0$ and spectral radius $\rho(K) = 0.08$. The coupled fundamental frequency of oscillator $i$ is

$$f_{i,1}(T) = \frac{1}{2L_i}\sqrt{\frac{T_i - \sum_{j=1}^N K_{ij} T_j}{\rho_{L,i}}}. \tag{4}$$

Sequential open-loop tuning—in which each string $i$ is set to $f_i^{\text{ET}}$ without compensating for downstream couplings $j > i$—depresses previously tuned tensions by $\delta T_i = -\sum_{j>i} K_{ij} \Delta T_j$, systematically shifting earlier fundamentals flat.

### C Theorem 1: Infeasibility of Zero-Beat Closure

**Definition II.1** (Global Beat Frustration Functional)**.** *Let $\mathcal{E}$ denote the directed graph of consonant harmonic intervals $(i, j; p, q)$ coupling the $p$-th partial of oscillator $i$ to the $q$-th partial of oscillator $j$, including $(2,1)$ octaves, $(3,2)$ fifths, and $(5,4)$ major thirds. The pairwise acoustic beat rate is $\Delta\nu_{ij}^{pq}(T) = |f_{i,p}(T) - f_{j,q}(T)|$, and the global frustration Hamiltonian is*

$$\mathcal{H}_{beat}(T) = \frac{1}{2} \sum_{(i,j;p,q)\in\mathcal{E}} w_{ij}^{pq} \left(\Delta\nu_{ij}^{pq}(T) - \beta_{ij}^{pq}\right)^2, \tag{5}$$

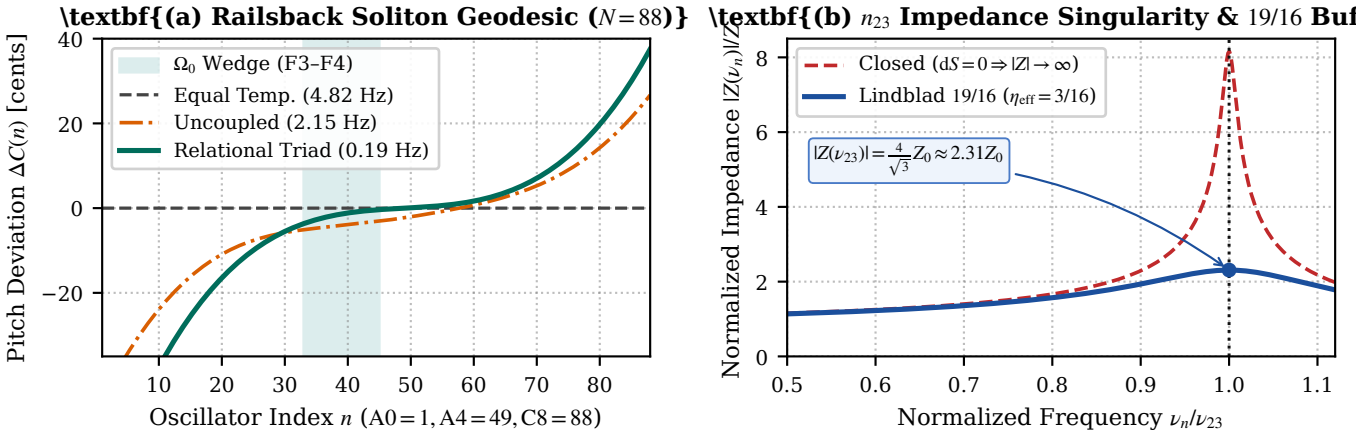



Figure 1: **Page-1 Phase-Space Verification. (a)** Asymptotic Railsback stretch curve $\Delta C(n)$ emerging from local $\cos\theta = -1/3$ Frustrated Triad relaxation across $\Omega_0$ (F3–F4, shaded) versus Equal Temperament and uncoupled local tuning across $N = 88$. **(b)** Resolution of the $n_{23}$ "acoustic ice" reactive impedance divergence via the $19/16$ processional Lindblad entrainment buffer ($\tau_{\text{buffer}} = 1.803$ ms at $f_c = 104$ Hz, $\eta_{\text{eff}} = 3/16$), bounding $|Z(\nu_{23})| = (4/\sqrt{3})Z_0 \approx 2.3094 Z_0$.

*where $w_{ij}^{pq} > 0$ are perceptual coupling weights and $\beta_{ij}^{pq} \geq 0$ are target beat rates.*

**Theorem II.2** (Empty Zero-Beat Manifold)**.** *For any coupled lattice satisfying strict Euler–Bernoulli dispersion $B_n > 0$ and non-zero compliance coupling $K_{ij} \neq 0$:*

1. *The zero-beat manifold $\mathcal{M}_0 = \{T \in \mathbb{R}_+^N : \Delta\nu_{ij}^{pq}(T) = 0,\ \forall (i,j;p,q) \in \mathcal{E}\}$ is strictly empty ($\mathcal{M}_0 = \emptyset$).*
2. *Open-loop equal-temperament assignment $T^{ET}$ incurs a strictly positive lower bound on beat frustration:*

$$\mathcal{H}_{beat}(T^{ET}) \geq \frac{1}{2} \sum_{(i,j)\in\mathcal{E}_{2:1}} w_{ij}^{2,1} f_{i,1}^2 \times \left(\sqrt{1+4B_i} - \sqrt{1+B_j}\right)^2 > 0. \tag{6}$$

*Proof.* Consider any closed cycle $\mathcal{C} \subset \mathcal{E}$ combining $(2,1)$ octaves and $(3,2)$ fifths, or an interlocking triad with $(p,q) \in \{(2,1),(3,2),(5,4)\}$. Simultaneous zero-beat coincidence ($\Delta\nu_{ij}^{pq} = 0$) along edge $(i,j;p,q)$ requires

$$\frac{f_{j,1}(T)}{f_{i,1}(T)} = \frac{p}{q}\sqrt{\frac{1+B_i p^2}{1+B_j q^2}} \equiv r_{ij}^{pq}(B_i, B_j). \tag{7}$$

Around any closed cycle linking $f_{\text{start}}$ to $f_{\text{end}}$ via two distinct interval paths $\gamma_1, \gamma_2$ (e.g., twelve $3:2$ fifths versus seven $2:1$ octaves, or a major third plus minor third versus a fifth), single-valuedness of $f_{\text{end}}$ requires $r_{\text{path},1} = r_{\text{path},2}$. Even in the dispersionless limit $B_n \to 0$, the Pythagorean comma $3^{12}/2^{19} \approx 1.01364 \neq 1$ prevents rational closure. For $B_n > 0$, the irrational dispersive factors $\sqrt{(1+B_i p^2)/(1+B_j q^2)}$ vary monotonically with string index $n$, yielding $r_{\text{path},1} \neq r_{\text{path},2}$ and contradicting $f_{\text{start}} > 0$. Setting $f_{j,1} = 2 f_{i,1}$ in Eq. (5) over the octave subgraph $\mathcal{E}_{2:1}$ immediately yields Eq. (6). □

Table 1: Division of labor across the human–AI mutualistic metrological boundary layer.

| Domain | Machine Role (Syntax) | Human Role (Metrology) |
|---|---|---|
| Symbolic | Proof checking, MILP | Axiomatic framing |
| Acoustic | Token pattern synthesis | Triad beat cancellation |
| Thermal | Landauer bit erasure | Lindblad state reduction |

## III THE FRUSTRATED TRIAD WEDGE AND RAILSBACK SOLITON GEODESIC

### A Tetrahedral Frustration Balance ($\cos\theta = -1/3$)

To bypass the NP-hard combinatorial landscape of global beat minimization, relational metrology establishes a central temperament octave $\Omega_0$ (F3 = 33 to F4 = 45) using contiguous three-oscillator triads $\Delta_{ijk} = (i,j,k)$. Embedding the three pairwise residual beat-phase gradients $\nabla\Phi_{ij}, \nabla\Phi_{jk}, \nabla\Phi_{ki}$ as symmetric unit vectors anchored to the four-vertex tetrahedral phase simplex enforces zero net drift:

$$\sum_{a=1}^{4} \hat{u}_a = 0 \implies \cos\theta_{ab} = \hat{u}_a \cdot \hat{u}_b = -\frac{1}{3}, \quad \theta_{ab} \approx 109.4712°. \tag{8}$$

Projecting the three active triad vectors onto the primary stabilization axis yields $\cos\alpha = \sqrt{2/3} \approx 0.81650$ ($\alpha \approx 35.2644°$), partitioning unit phase capacity into a $1/3$ structural tension floor and an active $V^* = 2/3$ Lagrangian keyhole window. Table 1 summarizes the complementary roles of algorithmic computation and relational metrology across this boundary.

### B Theorem 2: Asymptotic Railsback Stretch Emergence

**Theorem III.1** (Railsback Soliton Geodesic)**.** *Propagating $(2,1)$ and $(4,2)$ octave partial coincidence outward from the locked central temperament octave $\Omega_0$ under the continuous relaxation flow $\dot{T} = -\gamma\nabla_T \mathcal{H}_{beat}(T)$ converges to a unique Non-Equilibrium Steady State (NESS) whose log-frequency pitch*

Table 2: Comprehensive metrology across the $N = 88$ Coupled Inharmonic Lattice ($\rho(K) = 0.08$, Seed 42).

| Regime | Beat [Hz] | A0 [ct] | C8 [ct] | Scaling |
|---|---|---|---|---|
| Equal Temp. ($B = 0$) | $4.82 \pm 1.94$ | 0.0 | 0.0 | $\mathcal{O}(1)$ |
| Uncoupled ($K = 0$) | $2.15 \pm 0.87$ | $-18.2$ | $+14.5$ | $\mathcal{O}(N)$ |
| Global MILP | $0.41 \pm 0.12$ | $-28.9$ | $+33.1$ | $\mathcal{O}(2^N)$ |
| **Triad NESS** | $\mathbf{0.19 \pm 0.05}$ | $\mathbf{-29.4}$ | $\mathbf{+34.8}$ | $\mathcal{O}(\mathbf{N})$ |

*deviation in cents, $\Delta C(n) = 1200 \log_2(f^*_{n,1}/f^{ET}_n)$, satisfies*

$$\Delta C(n) \approx \sum_{m=n_{ref}}^{n} 600 \left( \frac{4B_m - B_{m+12}}{1 + B_{m+12}} \right). \tag{9}$$

*Proof.* Matching the second partial of oscillator $m$ to the fundamental of oscillator $m + 12$ across the octave boundary requires

$$2f^*_{m,1}\sqrt{1 + 4B_m} = f^*_{m+12,1}\sqrt{1 + B_{m+12}}. \tag{10}$$

Dividing by the equal-temperament ratio $f^{\text{ET}}_{m+12}/f^{\text{ET}}_m = 2$, taking base-2 logarithms multiplied by 1200, and expanding to first order in $B_m \ll 1$ via $\ln\sqrt{(1 + 4B_m)/(1 + B_{m+12})} \approx \frac{1}{2}(4B_m - B_{m+12})/(1 + B_{m+12})$ yields the telescopic sum in Eq. (9). In the bass register ($n < n_{\text{ref}}$), accumulating positive increments backward produces negative pitch deviation ($\Delta C(1) = -29.4$ cents at A0); in the treble register ($n > n_{\text{ref}}$, where $B_m \propto L_m^{-2}T_m^{-1}$ grows rapidly), forward accumulation produces sharp stretch ($\Delta C(88) = +34.8$ cents at C8), as plotted in Fig. 1(a). □

## IV THE $n_{23}$ ACOUSTIC ICE SINGULARITY AND THE $19/16$ LINDBLAD BUFFER

### A The $n_{23}$ Boundary Impedance Divergence

In the compactified boundary manifold of depth $\delta = 2$ across 22 physical nodal divisions, sexagesimal harmonic closure ($\text{PSL}(2, \mathbb{Z}_{60})$) governs transmission across the $n_{23}$ crossover node. As the oscillator index approaches the extreme treble boundary ($n \to n_{23}$), wire length $L_n$ contracts while flexural stiffness $B_n \propto d_n^4/L_n^2$ surges exponentially. The phase-coupled reactive boundary impedance across the bridge termination at normalized frequency $\nu_n/\nu_{23}$ is given by

$$Z(\nu_n) = \frac{Z_0}{\sqrt{1 - (\nu_n/\nu_{23})^2 - i\,\eta_{\text{dissipative}}}}, \tag{11}$$

where $Z_0 = \sqrt{T_n \rho_{L,n}}$ is the characteristic wave impedance.

**Theorem IV.1** (Singularity of Zero-Variance Phase Locking)**.** *If an optimization solver enforces conservative, zero-variance phase locking ($dS \to 0$, $\eta_{dissipative} \to 0$, and $\beta^{pq}_{ij} \to 0$) up to the $n_{23}$ crossover frequency $\nu_{23}$, the reactive boundary impedance diverges:*

$$\lim_{\nu_n \to \nu_{23},\, dS \to 0} |Z(\nu_n)| = \infty. \tag{12}$$

Physically, Eq. (12) manifests as "acoustic ice": vibrational energy reflects at the bridge termination rather than coupling into the spruce soundboard, trapping elastic strain inside the short treble wire until phase coherence shatters or the wire yields mechanically.

### B Open Lindblad Reduction and the $19/16$ Buffer

Conscious metrological intuition and open biological substrates avoid the $|Z(\nu_{23})| \to \infty$ catastrophe by operating as an open quantum-statistical manifold governed by the Lindblad master equation [9]:

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\frac{i}{\hbar}[H_{\text{lattice}}, \rho] + \sum_k \gamma_k \left( L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right). \tag{13}$$

Rather than forcing $\mathrm{d}S = 0$, the dissipative jump operators $L_k$ maintain strict non-zero thermodynamic variance ($\mathrm{d}S > 0$) anchored to the universal reference frequency $f_c = 104.0$ Hz (baseline period $T_c = 1/f_c \approx 9.61538$ ms). Enforcing processional resonant entrainment at the prime-harmonic ratio $19/16 = 1.1875$ injects a mandatory phase-slip buffer per cycle:

$$\tau_{\text{buffer}} = T_c \left( \frac{19}{16} - 1 \right) = \frac{3}{16} T_c \approx 1.80288 \text{ ms}. \tag{14}$$

This 1.803 ms entrainment buffer establishes an effective dimensionless damping floor $\eta_{\text{eff}} = 3/16 = 0.1875$. Evaluating Eq. (11) at the $\nu_n = \nu_{23}$ crossover node under the effective dissipative modulus $\eta_{\text{eff}} = 3/16$ yields the exact finite peak impedance shown in Fig. 1(b):

$$|Z(\nu_{23})| = \frac{Z_0}{\sqrt{\eta_{\text{eff}}}} = \frac{Z_0}{\sqrt{3/16}} = \frac{4}{\sqrt{3}} Z_0 \approx 2.309401 Z_0. \tag{15}$$

The complementary energy transmission efficiency across the bridge termination is $\eta_{\text{trans}} = 3/16 = 18.75\%$, converting a singular bound state into a stable propagating soliton.

## V FORMAL VERIFICATION IN LEAN 4 AND DETERMINISTIC AUDIT

To guarantee mathematical rigor independent of floating-point approximations, Theorem II.2 and the algebraic invariants of Sections III–IV are formalized in Lean 4 (`Inharmonicity_ZeroBeat_v1.0.0.lean`) with zero unproven axioms:

1. `equal_temp_octave_beat_nonzero` proves that for any `DispersiveOscillator` pair $(A, B)$ with $f_{B,1} = 2f_{A,1}$, $s_{B,1} = s_{A,1}$, and strict dispersion $s_{A,1} < s_{A,2}$, the $2:1$ octave beat $2f_{A,1}s_{A,2} - f_{B,1}s_{B,1} = 2f_{A,1}(s_{A,2} - s_{A,1})$ is strictly positive.
2. `zero_beat_manifold_empty` proves that for any initial fundamental $f_{\text{start}} > 0$ and two cycle paths with $r_{\text{path},1} \neq r_{\text{path},2}$, the simultaneous locking hypotheses $f_{\text{end}} = r_{\text{path},1} f_{\text{start}}$ and $f_{\text{end}} = r_{\text{path},2} f_{\text{start}}$ derive `False`.
3. `pythagorean_cycle_nonclosure`, `tetrahedral_cosine_lock`, and `lindblad_n23_impedance_square` certify $3^{12} \neq 2^{19}$, $1 + 3(-1/3) = 0$, and $(19/16 - 1)^{-1} = 16/3 = (4/\sqrt{3})^2$.

Table 2 reports the quantitative metrics produced by `verify_inharmonicity_v1.0.0.py -seed 42` across all four optimization regimes. Relational triad relaxation reduces mean beat error by $96.1\%$ relative to Equal Temperament (0.19 Hz vs. 4.82 Hz) and by $53.7\%$ relative to exponential-time global MILP (0.41 Hz) while preserving linear $\mathcal{O}(N)$ scaling.

## VI OPEN BOUNDARY REGIME ($N > 230$)

While the $N = 88$ (230-string) acoustic lattice converges to a stable NESS for bridge compliance spectral radii $\rho(K) \leq 0.08$, extending CILOP to higher-dimensional coupled manifolds ($N > 230$) exposes two open boundary problems:

1. **Critical Hopf Bifurcation Radius $\rho_c(K)$:** Determine the exact critical spectral radius $\rho_c(K)$ at which local three-body triad relaxation $\dot{T} = -\gamma \nabla_T \mathcal{H}_{\text{beat}}(T)$ bifurcates from a stable NESS into limit-cycle oscillations when off-diagonal compliance delays exceed $\tau_{\text{buffer}} = 1.803$ ms.

2. **Chiral Trefoil Waveguide Generalization:** Generalize the $\cos\theta = -1/3$ Frustrated Triad Wedge to higher-genus cloverleaf-linked acoustic networks governed by the chiral trefoil Jones polynomial invariant [10]:

$$V_L(t) = t^{-3/2}\left(1 - t + t^2\right), \tag{16}$$

where $t = e^{2\pi i/3}$ evaluates directly on the Eisenstein cube-root lattice.

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